

MAHIKA MAHAPATRA

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• Portfolio: mahikamahapatra101.github.io/Digital-Portfolio-v2

CS sophomore who ships. Built a production injury-risk tracker from 1,300+ real runs, and an LLM underwriting pipeline at CBRE. Comfortable taking projects from raw data to deployment.

EDUCATION

University of Wisconsin–Madison

Sep 2025 – May 2028 (Expected)

B.S. Computer Science | GPA: 3.40 / 4.00 | Dean's List, Spring 2026

Relevant coursework: Programming II & III / Data Structures (CS 300/400), Machine Organization & Programming (CS 354), Discrete Mathematics (CS 240), Linear Algebra (Math 340), Statistics (STAT 240), User Interfaces (CS 571)

TECHNICAL SKILLS

Languages: Java, JavaScript, HTML/CSS, Bash, R, Python, SQL, LC3, C

Tools & frameworks: Git/GitHub, React, FastAPI, SQLAlchemy, pandas, JUnit 5, VS Code, Eclipse, n8n, Supabase, PostgreSQL, Google Gemini / AI Agents

PROJECTS

RunLoad — Training Load & Injury-Risk Tracker

Jun – Jul 2026

Python · FastAPI · React · PostgreSQL · pandas | Full-Stack Web App | github.com/mahikamahapatra101/runload

- Built a full-stack web app (Python/FastAPI, React, SQLAlchemy) that calculates Acute:Chronic Workload Ratio (ACWR), a sports-science metric, to flag when a runner's training load is rising too fast.
- Wrote a Python pipeline to clean and process 1,300+ real runs pulled from a personal Strava export. The raw export didn't include perceived effort, so I added that field myself and reconciled it against the rest of the data manually.
- Hit a data loss bug after deploying to Render: SQLite doesn't persist on their free tier, wiping imported run history on restart.
- Migrated the backend to Postgres on Neon to fix it for good, rewriting the SQLAlchemy connection logic and re-importing all 1,300+ runs from scratch, then deployed the full app live on Render and Vercel.

ForecastGrid AI — Smart Grid Demand Forecasting

Jun 2026

R · LSTM Neural Network · Data Engineering · Time-Series Forecasting | FUSE Program, MKE Tech Hub Coalition | "In It To Win It" Award Winner

- Won the "In It To Win It" Award out of 6 competing teams at FUSE Demo Day hosted by the MKE Tech Hub Coalition.
- Merged three datasets that were never meant to line up: commercial energy, residential consumption, and live Milwaukee weather feeds, in R, writing custom alignment logic to fix mismatched timestamps that broke every join.
- Engineered features from the cleaned data and trained an LSTM model to predict energy demand spikes one hour ahead, giving operators a real warning window instead of a real-time surprise.
- Flagged 165 high-risk demand hours out of 1,643 hours of real Milwaukee-area grid data; built a dashboard prototype so control room operators could see a spike coming and act on it before it hit.

Stride — Campus Safety Navigation App

Mar 2026

Java · React · Dijkstra's Algorithm · Safety-Weighted Routing | CheeseHacks 24-Hour Hackathon | github.com/mahikamahapatra101/stride-cheesehacks

- Back-end engineer on a 4-person team: implemented Dijkstra's pathfinding from scratch (priority queue, visited sets, path reconstruction) over a hand-mapped campus graph of real locations and paths.
- Weighted the graph by lighting and foot traffic instead of raw distance, so the algorithm favors a longer, better-lit route over the shortest one. Adjusted the weighting multipliers through repeated testing, since too weak and it ignored safety entirely, too strong and it sent people on massive detours.
- Reworked the edge weighting mid-build after the first version produced routes that didn't make sense, retuned the multipliers under a hard 24-hour clock, and re-demoed a working version before time ran out.
- Engineered the back-end API for the React front-end and built a local routing fallback to ensure the app stayed fully operational during the live demo if cloud API credits failed.

Self-Balancing Search Tree

Jan – Mar 2026

Java · Red-Black Tree · Generics · JUnit 5 | github.com/mahikamahapatra101/SelfBalancingSearchTree_RedBlackTree_JUnit

- Implemented a generic Red-Black Tree with $O(\log n)$ guaranteed search and insertion, handling all rotation and color-adjustment cases from scratch with no external libraries.
- Verified all RBT invariants via JUnit 5 across arbitrary object types using Java generics; used as a game leaderboard backend supporting level, playtime, and record-count sorting.

EXPERIENCE & LEADERSHIP

CBRE & Project Destined — AI Engineering Fellow

June – Aug 2026

n8n · Google Gemini (AI Agent) · Lovable · Supabase · PostgreSQL | github.com/mahikamahapatra101/MultiGuard-AI

- Selected for an applied AI fellowship in partnership with CBRE, focusing on LLM workflow automation and commercial real estate underwriting.
- Built an n8n and Google Gemini workflow to parse complex multifamily Offering Memorandums, extracting core financial metrics (NOI, cap rate, occupancy, debt yield) with strict schema constraints.
- Coded a deterministic risk engine enforcing institutional investment thresholds (90% occupancy floor, 8% debt yield minimum) to automatically flag covenant breaches and output Go/No-Go verdicts.
- Connected a React frontend (built via Lovable) to a Supabase PostgreSQL database via secure webhooks to log deal audit trails and export PDF investment memos.

Ground Truth Cafe — UW–Madison | Barista

Jan 2026 – Present

- Handle high order volume and cash accuracy during peak campus rush while carrying a full CS courseload.

Science National Honor Society | Co-Founder, Sussex Hamilton High School

2024

- Founded the school's first-ever SNHS chapter: designed the application process, led a 6-month recruitment effort growing to 40+ members, and built all chapter programming with no existing template.

AWARDS & ACTIVITIES

- FUSE Program: "In It To Win It" Award Winner, MKE Tech Hub Coalition (2026)
- NCWIT Aspirations in Computing: State Winner 2025; State Honorable Mention 2022, 2023, 2024